SELF-IDENTIFYING THE MENTAL HEALTH STATUS AND GET GUIDANCE FOR SUPPORT

SUBMITTED IN PARTIAL FULFILLMENT FOR THE REQUIREMENT OF THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE

Submitted by

Prachi (2200290120120)
Pappu Teli (2200290120117)
Alok Kumar Gupta (2300290129001)
Yatendra Kumar(2300290129016)

Supervised by
(Bhagvan Krishna Gupta)
Assistant Professor

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DEPARTMENT OF COMPUTER SCIENCE

KIET GROUP OF INSTITUTIONS, GHAZIABAD

(Affiliated to Dr. A. P. J. Abdul Kalam Technical University, Lucknow, U.P., India)

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Signature:-

Name:- Prachi

Roll No.:- 2200290120120

Date:- 29/04/2026

Signature:-

Name:- Pappu Teli

Roll No.:- 2200290120117

Date:- 29/04/2026

Signature:-

Name:- Alok kumar Gupta

Roll No.:- 2300290129001

Date:- 29/04/2026

Signature:-

Name:- Yatendra Kumar

Roll No.:- 2300290129016

Date:- 29/04/2026

CERTIFICATE

This is to certify that Project Report entitled "Self- identifying the mental health status and get guidance for support" which is submitted by Prachi, Pappu Teli, Alok Kumar Gupta, Yatendra Kumar in partial fulfillment of the requirement for the award of degree B.Tech. in Department of Computer Science of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

.

Date: **29/04/2026** **Supervisor**

Mr. Bhagvan Krishna Gupta

(Assistant Professor)

Date: **29/04/2026** **Dr. Ajay Kumar Srivastava**

Dean

(Department of Computer Science)

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Signature:-	Signature:-
Name:- Prachi	Name:- Pappu Teli
Roll No.:- 2200290120120	Roll No.:- 2200290120117
Date:- 29/04/2026	Date:- 29/04/2026

Signature:-	Signature:-
Name:- Alok Kumar Gupta	Name:- Yatendra Kumar
Roll No.:- 2300290129001	Roll No.:- 2300290129016
Date:- 29/04/2026	Date:- 29/04/2026

ABSTRACT

During the last couple of years, mental health issues have increased, and the victims are usually youngsters and working professionals. Mental health challenges require early identification and timely support. This paper proposes a web-based platform for enabling users to self-identify their mental health status and receive specific guidance on various resources. The website introduces the user safely through the login system to a structured quiz on mental health based on medically validated psychological screening tools. At the end, a summarized analysis of the mental health status related to anxiety, depression, or stress is provided. Further, some lifestyle modifications, coping mechanisms, and professional mental health support are recommended based on the outcome. The location information of the candidates is going to be harnessed to give recommendations of qualified professionals in the mental health domain to the user. The user also has the ability to directly request an appointment with the system. Additionally, there is also going to be an online chatbot system that will sustain engagement and answer any of the often-asked questions by users of the website. The implementation of awareness in mental health issues is going to be achieved through the technological system available.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

ML – Machine Learning

MH – Mental Health

NLP - Natural Language Processing

SDLC – Software Development Life Cycle

UX – User Experience

DB – Database

NoSQL – Non-Relational Database

IDE – Integrated Development Environment

HTTP – Hypertext Transfer Protocol

CHAPTER 1: INTRODUCTION

1.1 Introduction

Psychological health is an important element of personal well-being today. In recent years, there has been a significant increase in mental health issues, including anxiety, depression and stress, affecting people of all ages and from all walks of life. This growth can be attributed to major life transitions, increasing educational and professional demands, a loss of social support and excessive digital media consumption. However, many people are unable to identify and address their mental health problems. This is due to a lack of awareness, stigma associated with seeking help and a lack of good initial screening tools.

Technology plays a crucial role in the delivery of health services, including mental health. Technology can be used to improve psychological support services to provide accessible, scalable, and user-friendly services, removing geographic, social and economic barriers. Digital mental health services offer an opportunity to screen for early signs of mental health problems. This allows people to investigate their mental health without fear of stigma or lack of services.

This new research describes how to develop a web-based mental health support tool for people with mental health issues. This online tool begins with a login system, allowing users to log in to their accounts while protecting and maintaining their privacy and confidentiality. These are crucial components of a mental health support tool. Once logged in, users are presented with a mental health assessment using psychological theories and diagnostic techniques. The assessment explores critical factors associated with stress, anxiety and depression. After completing the assessment, users get feedback and information on their mental health status. The tool also offers informative feedback in response to the results and make suggestions for enhancing their mental health, encouraging positive thinking. This could involve lifestyle modifications, stress reduction and other self-help methods.

It seeks to assist people in improving their self-awareness and mental health. To enhance the self-assessment and recommendations, the system also has a location service. This can provide location-specific recommendations for mental health professionals, such as psychologists and therapists. This facilitates access to services and appointments, linking the self-evaluation to support services. Apart from the initial assessment, there is also a chatbot service. This feature allows users to get information on-the-go about mental health problems.

The chatbot provides a virtual buddy to instigate users to continue to engage with their mental health status. The key objective of this project is to bridge the gap between mental health awareness and support services for people. The solution will be a safe, convenient and private system that allows for early detection, self-awareness and intervention. This research seeks to use technology to democratise mental health services and enabling access without fear of stigma.

1.2 Project Category

The proposed project for AI-Based Mental Health Support System is a "Health-care Informatics and Artificial Intelligence Application". The proposed project is a multidisciplinary research area that involves the incorporation of the principles learned from Artificial Intelligence, Machine Learning, Natural Language Processing (NLP), Cloud Computing and Web Technologies to address real-world problems in the domain of mental health awareness, assessment and online support.

The project is also a technology project under the topic "Intelligent Healthcare System". This system aims to improve the access, early detection and participation in health care through smart computing. The proposed system makes use of machine learning techniques for mental health screening and AI chatbots for emotional support, making it an example of the use of AI in health care.

Finally, the proposed system can be seen as an application of real-time online health monitoring and support. The proposed system provides users with mental health assessment by taking interactive quizzes, gives them recommendations based on the quizzes and allows them to chat with an AI chatbot for emotional support. The use of cloud-based databases enables it to be a real-time data storage, synchronization, and accessible platform.

From an academic standpoint, the project can be categorized as an AI-based research solution for the healthcare industry. The project highlights the use of AI models for the analysis of behavioral health and the importance of early mental health screening using technology. The project provides a solution to the conventional manual screening process through an automated intelligent analysis system, thus making it a software-based solution for the healthcare industry.

From an industry standpoint, the project can be categorized under the following categories: Digital Health and Telemedicine Solution. The project helps in the creation of an advanced healthcare system by highlighting the significance of preventive mental healthcare, without the need for physical contact for early screening, and reaching out to those who are resistant to traditional therapy.

Therefore, the proposed Mental Health Support System can be generally categorized as an AI-based Digital Healthcare Application that highlights the significance of early detection, user engagement, intelligent recommendations, and cloud-integrated mental wellness support.

1.3 Objectives

The key goals of the project are:

- To design an intelligent mental health support system.
- To achieve early detection of mental health issues.
- To enhance mental health awareness and accessibility.
- To provide personalized mental wellness recommendations.

The goals can be further extended in the following way:

Development of a Centralized Mental Health Platform

The proposed system will aim at developing a centralized platform for users to evaluate their mental health, get advice, and interact with AI tools. This will ensure that mental health support is organized.

Early Mental Health Assessment

The early detection of mental health problems like anxiety, depression, and stress through AI-assisted quizzes and analysis software is one of the primary objectives of the proposed system.

AI-Powered Personalized Recommendations

The application provides personalized recommendations such as lifestyle modifications, coping Strategies, and wellness practices based on the mental health assessment result of the user.

Intelligent Chatbot Support

An AI-based chatbot is integrated to provide emotional support, basic support, and instant interaction, which helps in enhancing user engagement and usability.

Doctor Recommendation and Appointment Support

The application assists users in getting in touch with mental health professionals by recommending doctors and scheduling appointments whenever professional help is required.

Enhanced Accessibility to Mental Health Care

The application addresses issues such as social stigma, unawareness, and geographical barriers by providing a remote mental health solution that can be accessed from anywhere.

User-Friendly Experience

The web application will be intuitive and interactive to ensure the convenience of navigating mental health tests.

Data-Driven Insights and Monitoring

The system will be able to monitor user input and provide information that could improve future mental health predictions and suggestions.

Scalability and Future Improvements

The design of the app can consider future aspects in terms of new AI models, mobile app, multi-lingual support and real-time monitoring of therapy.

All of this considered, the goal of this project is to offer an intelligent, scalable approach to mental health.

1.4 Structure of Report

The report will be split into several chapters covering the particular phases of the development process.

Chapter 1: Introduction

The chapter will include information on the problematics of mental health conditions, causes of the project, project goals, and characteristics of the project.

Chapter 2: Literature Review

The chapter will include an overview of current technological solutions for mental health treatment, AI healthcare solutions, chatbot platforms, and wellness applications. It will highlight the weaknesses of these solutions and gaps in the literature.

Chapter 3: Proposed System

The proposed system design along with the AI models and chatbot implementation is presented in this chapter.

Chapter 4: Requirement Analysis and System Specification

Feasibility analysis, software requirements, design specification of the system, and database design are presented in this chapter.

Chapter 5: Implementation

Tools and technologies used, development environment, dataset implementation, and integration of AI models into the web application are presented in this chapter.

Chapter 6: Testing and Maintenance

Testing strategies for the proposed system, usability testing, performance testing, and future approaches to maintenance are presented in this chapter.

Chapter 7: Results and Discussions

Experimental results, user evaluation parameters, accuracy of the system, usability test results, and real-world experiment analysis are presented in this chapter.

Chapter 8: Conclusion and Future Scope

Conclusion of the entire project work and future scope of the project are discussed in this chapter. Future scope of the project includes mobile-based implementation, AI therapy bots, and mental health analytics.

CHAPTER 2: LITERATURE REVIEW

2.1 Literature Review

Mental health has emerged as a serious concern for the global community in recent years due to the increasing number of cases of stress, anxiety, and depression, especially among students and working professionals. With the advancements in technology, there has been a growing interest in the development of various digital solutions for improving mental health awareness, screening, and support systems. Mental health care technologies have evolved from traditional clinical tools to intelligent AI-based digital solutions in recent years. This section provides a comprehensive review of the various approaches and technologies used in mental health support systems.

2.1.1 Self-Assessment and Psychological Screening Systems

Among the early studies on self-identification of mental health issues is the study carried out by Evans-Lacko et al. (2019), which developed the SELF-I (Self-Identification of Having a Mental Illness) scale. The study highlighted the importance of self-identification of mental health issues in seeking professional help. The results of the study highlighted the importance of having a structured self-screening system that can help reduce stigma and promote help-seeking.

The PHQ-9 and GAD-7 questionnaires are also used as traditional screening tools to measure the levels of depression and anxiety. However, most of the screening tools are not entirely integrated with web-based platforms and require the monitoring of mental health professionals. This has led to the development of automated web-based mental health assessment systems.

2.1.2 AI-Based Chatbots in Mental Health

Conversational agents have also been increasingly used as a supportive system for mental health. A supportive system is Woebot, which was used in Fitzpatrick et al. (2017). The findings revealed that the use of chatbot conversations based on AI and CBT approaches led to a significant reduction in symptoms of anxiety and depression.

Mobile applications like Wysa, Youper, and Moodpath also provide conversational support, mood tracking, and self-help exercises. These applications are developed using natural language processing and pre-defined therapeutic protocols.

Even though chatbot systems are highly beneficial in terms of accessibility and engagement, most of the applications available in the market either provide emotional support or symptom tracking, but not a comprehensive solution that includes assessment, expert advice, and appointment scheduling.

2.1.3 Location-Based Mental Health Services

Location-based digital mental health applications have been proposed for connecting users with mental health professionals in the location. A study was conducted by Luxton et al. (2011) to examine mobile health applications that used smartphone technology to provide location-based mental health information and crisis response.

Though location-based services improve accessibility to mental health professionals, the applications are not linked to intelligent assessment tools. They are only directory applications and not predictive applications.

2.1.4 Machine Learning in Mental Health Predictions

Machine learning has been increasingly used in mental health predictions using behavioral and psychological factors. Based on the studies written and published in scholarly journals such as Psychological Medicine and NPJ Digital Medicine, there are many research papers that focus on using supervised machine learning algorithms such as logistic regression, decision trees, and random forest in predicting mental illnesses.

The following parameters are considered when using the algorithms:

- Age
- Sleep patterns
- Screen time
- Work pressure
- Anxiety scores
- Depression scores

A research paper authored by Torous et al. (2021) emphasizes the role of implementing artificial intelligence in increasing access to mental health interventions through digital media.

Despite the promising findings from the application of the algorithms, the algorithms lack good user interface webpages where users can get accurate predictions.

2.1.5 Digital Mental Health Platforms

Technology has become essential for developing efficient mental health intervention systems in the world, as stated by WHO(2022). There are both mobile and web-based applications that address the issue of increasing awareness to seeking professional help.

However, mental health intervention tools currently available online can be classified as follows:

- Symptom tracking applications
- Online therapy booking systems
- AI-based chatbot systems
- Mental health awareness portals

Very few systems combine all the above aspects into a single intelligent, secure, and scalable web-based system.

2.2 Research Gaps

Although a great deal of progress has been achieved in digital mental health solutions, there are still some research gaps that need to be filled. Conventional mental health solutions are highly dependent on manual diagnosis and are not scalable for large groups of people. Mobile mental health interventions are accurate but very inefficient. AI mental health interventions are structured but emotional.

Apart from the aspect of being separate models for prediction purposes and not being integrated into any software application, the current machine learning techniques lack integration of all components including self-evaluation, AI support, chatbots, and expert guidance.

Another major drawback is that most solutions do not provide personalized mental health advice based on individual behavioral patterns. Moreover, most solutions do not provide secure authentication, structured user profiling, or long-term monitoring support.

2.3 Problem Formulation

The increasing number of mental health issues, as well as the lack of accessible early screening systems, has resulted in a huge need for intelligent digital solutions. Most people are not in a position to detect the early signs of mental health issues because of the stigma that is associated with mental health issues, lack of awareness, and lack of access to mental health services. Most people normally wait until the issue is beyond control before seeking assistance, and this is why early detection and accessible digital intervention platforms are vital.

The problem statement can therefore be argued to be the need to create an AI-powered web-based mental health intervention platform where people are able to assess their mental health issues and receive professional advice when needed.

The system comprises a variety of major components that interact with each other to provide a complete solution. The components of the system include a secure user authentication component for privacy and security of data, an AI-based mental health assessment component in the form of an interactive quiz, and a machine learning analysis component for analyzing the responses of the user. The system also comprises a personalized recommendation engine component for providing personalized recommendations, an AI chatbot interaction component for real-time interaction, a cloud database component for secure storage, and a web interface component along with an administrative interface for monitoring the system.

The functioning of the system begins with user registration and secure login into the system. After the user has been able to log into the system, the next process involves them being required to undergo an assessment quiz on their mental health. The quiz will assess how stressed, anxious, or depressed the user feels. The machine learning algorithms will then help analyze the answers provided by the user to know whether they are stressed, anxious, or depressed. On completion of the assessment and analysis of data, the machine learning algorithm will provide personalized advice and coping mechanisms to enable the users cope with their mental state.

Another user-friendly approach included in this project is the provision of an AI chatbot that can be used to answer most common questions on mental health. In addition to this, the user will be able to make an appointment and see a mental health expert whenever they wish to.

The major objective of this project is to help facilitate early detection of any mental health problem and provision of digital mental health services. Through early detection, the system will ensure that patients do not shy away from mental health screening due to stigma.

CHAPTER 3: PROPOSED SYSTEM

3.1 Proposed System

The proposed system is an artificial intelligence-based web-based mental health support system that seeks to provide early detection, recommendations and mental health care. The proposed system takes advantage of the power of artificial intelligence, machine learning, and web technology to provide a safe and user-friendly system for mental health promotion and intervention.

The proposed system is a web-based platform which allows users to assess their mental health status and receive recommendations without the need for face-to-face interaction, unlike a traditional mental health support system. The proposed system design is modular and scalable to allow the incorporation of smart analysis, cloud computing, and user interfaces. The proposed system design comprises layers performing the functions of the system such as user authentication, mental health testing, analysis, recommendation, chatbot interactions, and monitoring.

3.1.1 User Authentication Layer

User authentication will guarantee the access of the users to the system while securing the confidentiality of their mental health information. User authentication will involve user registration, user login, and user profiles. Security measures will be employed in designing the user authentication layer in order to enhance the confidentiality and privacy of the information. After logging into the system, the user will see their own personalized dashboard which contains test results and other past sessions.

3.1.2 Mental Health Assessment Module

The mental health assessment module is the primary input module of the system. The mental health assessment module is a scientifically designed and structured quiz that assesses stress, anxiety and depression. This quiz is formulated using valid psychological factors and behavioral aspects like sleep patterns, work stress, mood and lifestyle. This fun quiz is a good way to guide the user in contemplating their mental health while generating meaningful input data.

3.1.3 Machine Learning Analysis Layer

The machine learning analysis layer is the part of the system that processes the user data and predicts the mental health. The user data is analysed by using different supervised machine learning techniques such as Logistic Regression, Decision Trees, and Random Forest Classifiers.

These models are based on the preprocessed datasets that have attributes related to users' demographic, behavioral and mental health characteristics. The analysis layer also aims to find the relationships between attributes such as anxiety, depression, sleep duration and screen usage to make predictions.

3.1.4 Personalized Recommendation Engine

The machine learning analysis layer is responsible for the analysis of the user's responses and predicting the mental health status. The analysis layer uses several supervised machine learning techniques such as Logistic Regression, Decision Trees and Random Forest Classifiers.

The classifiers are trained on the preprocessed datasets that have attributes related to the demography, behaviour and psychology of the users. The analysis layer also tries to establish relationships between the attributes such as anxiety, depression, screen time and sleep time to make accurate predictions.

3.1.5 AI Chatbot Interaction Module

To make sure that the users remain motivated and engaged and to provide them with emotional support, the system uses an AI-based chatbot. The chatbot is developed with natural language processing and it is trained on mental health-related conversations.

The chatbot helps the users in responding to their general questions, providing motivational support and helping the users navigate through the different functions of the app. The chatbot engages the users through the instant feedback, reduces isolation and supports the users to seek help when required.This is an important feature in the post-assessment engagement of the user.

3.1.6 Doctor Recommendation and Appointment Module

The system includes a location-based doctor recommendation module that provides access to mental health experts. Based on user preferences and needs, the system will provide a list of local therapists/counselors, their details, and an option of online communication with the doctor.

The system provides access to the professionals' profiles and appointment scheduling. The module facilitates the connection between online screening and professional mental health services by helping users who want to seek professional help to do so.

3.1.7 Web Interface Layer

The web interface layer provides the user with an easy and responsive interface. The interface is developed using the most recent web technologies to ensure that the interface can be accessed on a range of devices such as computers, tablets and phones.

The system is designed with a simple and user-friendly interface to make mental health services available to people of all ages. The interface is developed with dynamic rendering to ensure that there are real-time updates, and users can instantly access their results.

3.2 Working Process of the System

The working of the proposed system is completely automatic and easy to use. The users register and securely log in to the system. Next, they will be required to complete a mental health questionnaire. Then the machine learning algorithms analyse the answers and predict the state of the user's mental health.

Based on the prediction, the system offers the user with personalized suggestions. In addition, users can also chat with the AI chatbot for immediate support. If necessary, the system also recommends mental health experts and allows the users to book appointments. The users' data is stored in the cloud database, which enables instant access and support.

3.3 Unique Features of the Proposed System

There are some distinctive features of the proposed system, in comparison to conventional mental health systems. The most notable feature of the proposed system is that it uses machine learning for mental health predictions. This can be particularly helpful for early detection and scaling. The addition of the AI chatbot feature leads to better user engagement and support.

The proposed system integrates assessment, recommendations and networking. Another key aspect of the proposed system is that it is secure and private with secure user authentication and encryption of data. The proposed system is scalable due to a cloud-based approach. This will benefit future system enhancements including mobile app connectivity, language translation and new AI-powered therapeutic tools. The proposed system is an advanced and user-friendly digital solution for mental health support, with its smart analytics and user-friendly nature.

CHAPTER 4: Implementation and Results

4.1 Feasibility Study

The proposed AI-based web mental health support system has been evaluated for its technical, economic and operational feasibility in the feasibility study. This will help to assess the feasibility of the proposed system to be successfully implemented with currently available technology, at a reasonable cost, and with effective use by the intended users. Given the rise in the need for accessible online mental health solutions, the system has been designed to be scalable, cost-effective and technically feasible using existing web and artificial intelligence technologies.

From a technological perspective, the proposed system has been built as a full-stack web-based application. The user interface of the proposed system has been designed using HTML, CSS, and JavaScript, which allows a neat and responsive interface for the users. These technologies provide support to multiple platforms such as desktop, tablet, and smartphones.

The back-end of the proposed system has been designed using Python-based frameworks, which is suitable for integrating machine learning models and server side operations such as user management, data processing and predictions. The modular design allows for the separate development and integration of system components including the assessment, chatbot, recommendation engine and admin dashboard. The feasibility study also examined issues related to deployment and scalability. The web-based nature of the system makes it feasible to deploy the system in the cloud. This will ensure that the system can be scaled on the basis of users.

The use of cloud services for the system will ensure the system has high availability, data backup and remote access. This will ensure that the system is scalable for deployment. The cloud services will also ensure that the system can store user's sensitive data and has the required performance and reliability. This will mean that future scalability is not required and costly. The practicality of the system, in terms of security and privacy, was also a key consideration, as the system will be dealing with sensitive data related to mental health.

The system has been developed with secure authentication, encryption and access control mechanisms. This will ensure the confidentiality and security of the user data. From a usability point of view, the system has been developed for non-technical users. The interface design is simple and intuitive to navigate through the system to access the mental health assessment, chatbot and view the recommendations. The inclusion of the AI chatbot functionality enhances the system's usability, which is beneficial to the users who are unwilling to use the traditional mental health services.

Overall, the feasibility study has demonstrated that the AI mental health support system is technically feasible and implementable. The availability of modern development tools, open source AI tools and cloud support make it feasible and sustainable for implementation. The system provides a good balance between innovation and feasibility to be implemented as a digital system for mental health support.

4.1.1 Economic Feasibility

The proposed AI-Based Web Mental Health Support System is cost-effective as it uses open-source technology and cheap software development tools. The tools required to develop the system, such as machine learning libraries built on Python, web development and cloud-based database solutions, are free or have a low monthly subscription fee. This significantly affects the cost required in developing the system and thus makes it affordable for use in educational institutions, startups, hospitals, and individual level use.

The system is also not very hardware intensive as it can be developed on normal computers, laptops, or even on the cloud without any special hardware. The system is web-based and so a user would only need an internet-connected device such as a mobile phone or a computer in order to access the services.

The system uses machine learning models that are efficiently developed on moderate computational hardware and don't require special hardware. Maintenance costs are also low as upgrades, improvements and development can be done via software updates instead of hardware replacements. This implies that the chatbot knowledge base, recommendation system and the machine learning models can be updated through software, thus ensuring that the system is sustainable in the long term and the operational costs are manageable.

The system can be sustainably cost-effective in the long term due to its many benefits. This is because the system results in the early detection of mental health issues, avoiding the need for serious treatment and emergency care. This leads to the system being able to provide benefits in terms of productivity gains in the workplace and institutions, and reduced healthcare costs.

Conclusion

In summary, the fact that the system has low initial costs, low hardware costs, low maintenance costs and the socio-economic benefits in the long term means that the proposed AI-based mental health support system is economically sustainable and cost-effective.

Cost Factor	Impact
Software Licensing	Minimal (Open-source)
Development Cost	Moderate
Maintenance Cost	Low
Infrastructure Cost	Scalable & Controlled
Long-term Operational Savings	High

4.1.2 User Authentication and Security

Authentication of the users and data security is one of the most important components of the proposed AI-based mental health support system because the system involves highly sensitive psychological and personal data. To provide confidentiality, integrity, and secure access, a secure authentication system was designed and implemented for the system. The authentication system provides secure user registration, login, profile management, and session management. While registering a user, the basic authentication details are collected, and they are stored securely in the database.

To ensure that unauthorized access is not made and user credentials are secure, password security mechanisms were developed using hashing algorithms. Rather than storing passwords in their original form, the system stores the encrypted hash values of passwords before they are stored in the database. This ensures that even if the database is compromised, it is impossible to retrieve the original passwords. Other security mechanisms, such as input validation and protection against common vulnerabilities , were also incorporated to further improve the integrity of the system.

The system also employs session-based authentication to facilitate the safe processing of user interactions. After the user has successfully logged in, a secure session token is created and sustained throughout the user session. This ensures that only the right people access the customized services that include mental health evaluation, recommendation by the AI, chatbot, and appointment booking. Expired timeouts will help automatically log out inactive users so that other persons cannot use the system on public or shared terminals to gain access into the system.

With regard to the confidentiality of the mental health records, safe communication techniques have been incorporated to ensure safe transfer of information between the client and server systems. Also, access controls ensure that unapproved individuals do not gain access to the system, thus making sure there are no information breaches at all.

Implementation of effective access controls increases the authenticity and integrity of the whole system since the information involved is sensitive in nature.

4.1.3 Operational Feasibility

The The practicality and feasibility of the proposed web-based mental health support system with AI technology is very high since it is used and operated by people who have less technical knowledge. The system is designed to be used by students, professionals and general users who need mental health services.

The web-based system is designed to be user-friendly and easy to navigate, with instructions and guidance to help users easily access the system components, such as mental health assessment, chatbot, and recommendations, even without extensive training. The system is also well integrated with the users' lives as it can be accessed using standard web browsers on mobile devices, tablets and computers.

The system does not require installation and users can simply use the system with internet connectivity. The automatic assessment and machine learning analysis of mental health reduce the amount of processing required, enabling users to quickly learn about their mental health. The system's recommendations are effective coping strategies and self-help techniques, which also enhance user acceptance and satisfaction.

The system makes sure it is beneficial and easy to use. The system also provides authentication and data protection capabilities, which also increase user acceptance and trust. Given that mental health data is regarded as sensitive data, it is crucial to ensure confidentiality and secure access to data. By putting in place measures that ensure that the system has robust data protection policies, the system increases user acceptance.

The management features also enable the moderators of the system to monitor and provide suggestions without compromising the system's performance. The cloud-based and modular system can be easily maintained and upgraded. New features like more advanced machine learning algorithms, knowledge bases for the chatbots, and other mental health resources can be added without compromising access.

All in all, the designed system appears quite feasible as it is not only easy to operate but does not require any specific skills for using. It is also user-friendly and secured with a solid infrastructure. All of these factors contribute to making the proposed design the viable digital mental health support system.

Table 4.1.4 Operational Feasibility Analysis

Aspect	Evaluation
User Acceptance	High
Ease of Use	High
Training Requirement	Minimal
Compatibility with Existing Process	High
Reliability	High

4.2 Software Requirement Specification

The proposed web-based AI-powered mental health support system requires some software for development, deployment, and maintenance. The user interface of the proposed system is built using HTML, CSS and JavaScript to create a responsive application.

The back-end system is developed using Python as it has a lot of support for web development and machine learning. The machine learning libraries are used to preprocess the data, design the models, and predict the presence of mental health problems like stress, anxiety and depression from user responses. The user data, answers to the user assessment questions, chatbot dialogs and prediction are stored in a database such as MongoDB or Firebase which is cloud compatible and scalable.

The proposed system can be built using popular development software and integrated development environments (IDEs) on the Windows, Linux or macOS operating systems. Further, a web browser is needed to interact with the system. The combination of all these software components enable seamless operation, scalability, security and reliability of the proposed mental health support system.

4.2.1 Data Requirements

The proposed web-based mental health support system using AI technology requires access to mental health data that is structured in a format that can be used for analysis and prediction. The proposed system uses data on responses to psychological assessment, behaviour, emotional expression, stress, lifestyle, and demographic data like age, gender and occupation of the user.

These data used in the proposed system can be used to detect patterns and risk factors for common mental health disorders such as stress, anxiety and depression. The proposed system employs historical and real-time data from the users' answers to the mental health quizzes, which are entered into the web-based system using interactive mental health assessment quizzes. Users need to provide answers to the structured mental health quizzes, which are analysed by the machine learning algorithm.

The data is preprocessed to clean, standardise and manage missing data to ensure that the results are accurate, consistent and reliable. The database stores several types of information, including user profile data, encrypted user authentication data, user's assessment responses, prediction results, chatbot interaction messages, and progress records. Given the critical nature of mental health information, the system implements secure data storage, access control and encryption to ensure that the information is not accessible by unauthorized persons.

The data and authentication are encrypted to ensure privacy and ethical considerations. Machine learning algorithms are heavily dependent on the quality, variety and volume of the dataset. Thus, this system can potentially be upgraded on a regular basis with a better dataset to provide more accurate predictions and also to keep up with the latest trend in psychological and behavioural patterns. By regularly updating the dataset, the system can learn and improve and offer more relevant data to the users. The system needs quantitative and qualitative data to be able to provide a holistic assessment of mental health status.

The quantitative data is test and frequency-based usage data and the qualitative data is descriptive feedback from the user's chat sessions with chatbot or self-assessment questionnaires. The use of these two data sets increases the prediction model's accuracy and effectiveness. The data collection is systematic and standardized for validity and reliability. The questionnaires used for assessment are standardized and validated based on psychological factors to ensure that the questionnaire does not miss the important indicators of mental health status.

Regular reassessment is made in order to enable the system monitoring changes and detecting the early signs of decompensation in one’s mental health condition. The data storage, management, and processing take place within a cloud database management system which allows for efficient data retrieval and analysis. Data structuring techniques, such as indexing, classification, and time-stamping, are also employed as part of the data management and storage strategy. Moreover, data backup and restoration are also included in the process of maintaining data integrity and availability. However, the issue of ethics should also be taken into consideration while managing data needs.

The system asks for user consent before the data is uploaded. Secondly, the system is transparent on the use of information. In cases where information is required, it is anonymized to prevent the user's identity from being disclosed. Adherence to data protection and privacy policies helps to guarantee the enhanced trustworthiness and confidence in the system. Furthermore, the system also provides data analytics features to develop insights and visualisations of mental health trends.

The data analytics feature of the system helps users to analyse their mental health patterns and make decisions on how to manage their mental health or seek professional assistance. Thus, data management, handling and processing play an important role in the successful design and development of the proposed AI-based mental health support system.

4.2.2 Functional Requirements

The web-based mental health support system using AI should have secure user login and registration functionality to allow users to securely login to the system. Users should be able to register, login, change their profile and manage their accounts securely. The system should include a role-based access control system that allows different access for users and administrators.

General users should be able to access the evaluation tools, the chatbot interactions and the support suggestions, while administrators should be able to access the system monitoring, dataset and user management. The system should enable users to complete interactive quizzes for mental health assessment to find out the symptoms of stress, anxiety and depression. The quizzes should be interactive and attractive to users.

Once the users have completed the quizzes and entered the answers, the system should process the input data using machine learning algorithms developed in the system to make prediction on mental health status. The system should provide the prediction results in real-time and present them in a user-friendly way. Another crucial requirement of the system is that it should provide personalised recommendations for mental health problems. It should offer coping skills, lifestyle tips, relaxation strategies and self-help tips based on the user's results.

The system should also provide on-the-spot support from the AI chatbot, answer questions about mental health and provide information to the users. The chatbot should also be available 24 hours a day to enhance user experience. The system should also be able to store user data, such as profile, assessment responses, prediction results, chatbot conversations and progress. The system should support users to view their previous assessment results, which can be used to track their mental health status.

The system should support the generation of basic dashboards or reports to visualise past information or progress made. Recommendation of appointments should also be implemented, which will enable users to get recommendations for mental health appointments when risky statuses are identified. Simple appointment or contact management should also be provided to seek professional help. Administrative features are also necessary.

The administrators should be able to manage user accounts, update assessment questionnaire, monitor system usage, manage dataset, and monitor the performance of the prediction models. The system should maintain efficient data retrieval, update and deletion functions with the proper access control. The above functional requirements ensure that the system can carry out the required operations such as secure login, mental health check, AI-based prediction, personalised help generator, chatbots, data storage and administration, etc.

4.2.3 Performance Requirement

The proposed web-based mental health support system employing AI technology should be able to provide accurate, authentic, and prompt results for effective digital mental health screening and support. The system should utilize standardized and validated psychological assessment questionnaires, such as stress, anxiety, and depression screening scales, to provide authentic results for assessment. The machine learning algorithm employed in the system should be able to provide high-accuracy prediction results with minimized processing time, allowing the user to obtain instant feedback after completing the assessment.

The system should be able to support multiple users simultaneously without hampering the performance of the system. As the proposed system is web-based, it should be able to provide scalability to support increasing user traffic, especially during peak usage hours. The cloud-based system should be able to provide high availability, fast response time, and minimized downtime. The response time for submitting the assessment and obtaining predictions should be a few seconds to retain user engagement and system efficiency.

It should be possible for the system to offer monitoring services in an uninterrupted manner since it will store historic assessment data and allow users to track any changes in the status of their mental wellbeing. Performance improvement measures like database indexing, use of caching, and effective API communication can help with this objective.

Confidentiality of the data and security of the system are other important considerations due to the sensitive nature of mental health information. It should be guaranteed that the data is transmitted securely via encrypted communications and that it is stored in a manner which ensures system safety. All relevant laws and policies regarding data privacy must be adhered to. The authentication and authorization processes of the application should prevent unauthorized access without compromising the system's responsiveness.

In addition, the application should function reliably and stably under normal circumstances. Data integrity and system functionality should be ensured by the presence of backup and recovery systems. In general, the system should be accurate, secure, scalable, responsive, and ethical, thus providing a trustworthy digital mental health support system connected with relevant self-help information and professional referral systems when required.

4.2.4 Maintainability Requirement

The AI-based web-mental health support system should be highly maintainable to ensure its continuous use and flexibility. The system should be built in a modular manner, such that each component of the system, such as the frontend design, machine learning module, chatbot engine, authentication module and database module can be individually updated or changed. This will allow the developers to modify the system without modifying the whole system, thus enabling easy maintenance of the system. The system should allow easy updating or modification of machine learning models, where the developers will be able to update or replace models with new data. Data and patterns related to mental health keep evolving and hence the system should be updated regularly to keep it up-to-date and accurate in its predictions.

The system should allow easy updating of models without impacting access to the system. The recommendation engine and chatbot database should also be easily updated to add new mental health tips and approaches. Maintainability will be accomplished by documenting code and having version control systems. Code documentation, API documentation, database documentation, and business logic documentation will provide easy maintainability of the system. Version control systems will provide change management, backup services, and collaboration to offer easy maintainability of the system. The system will integrate easily with the existing development platforms for easy maintenance of the system. Working with current development platforms will be convenient for maintaining the system. Working with popular development platforms such as web development frameworks and cloud-based databases will offer easy maintainability.

Deploying the system on the cloud will make it easy to maintain the system using popular development platforms. It will become easier to update the system, monitor the system, and scale the system using the cloud environment. Logging and error handling will be included in the system design to achieve easy error detection and maintain the system. Automatic logging of errors will be included to detect any type of error, including performance and security errors within the system. Backups and restoration of data will be included to maintain and protect the data.

4.2.5 Security Requirement

For the proposed web-based mental health support system using AI technology, it would be vital to integrate highly reliable security in the platform because of the sensitive nature of the data that the system will handle. Given that the proposed system will involve mental health data, it would be crucial to introduce high-level security features to enhance the confidentiality of such information to avoid any potential misuse. Some of the key security features for this purpose include secure log-in mechanism with password storage and role-based access to allow users access to specific data.

In addition, another security aspect that would play an important role in making sure that users' data is safe is data confidentiality. Since the proposed system involves mental health data, chatbot interaction and user data, it would be necessary to employ some encryption technologies for securing data both during its communication and storage. In addition, there would be need for ensuring data transmission security by integrating HTTPS and SSL/TLS in the application.

The system must include safeguards to counter threats such as attacks through unauthorized access, modification of data, injections, and session hijacking. Input validation, API control, and management of authentication tokens must be incorporated into the system to avoid risks in these areas. Session management must also be included to safeguard users using the system on public computers from risks resulting from shared computers where idle sessions remain open.

Data integrity is another important issue that must be considered. Data validation must be put in place to ensure that the data collected is genuine and has not been manipulated in any way. Changes in the system will also be tracked using access logs and audit trails. Data backup must be incorporated to avoid losing any data due to cyber attacks or deletion.

Privacy and ethical considerations must also be put in place during the design process. Users' consent must be sought before collecting their personal information. The system should also ensure that the users' information is not used for any other reason other than the evaluation of mental health and provision of support.

Additionally, logging and monitoring software should also be incorporated to monitor system events and alert of any unusual activities in real-time. Security monitoring helps the system administrator to be aware of any possible threats and take appropriate actions to mitigate them.

In conclusion, the above security features ensure that the proposed system has the highest level of confidentiality, integrity, and availability. The proposed system will therefore offer a safe and secure delivery of digital mental health support services through the adoption of proper authentication, encrypted communication, secure storage, and ethical handling of data.

4.3 SDLC Model Used

The proposed AI-based web-based mental health support system was developed by following a structured Software Development Life Cycle (SDLC) approach. The SDLC model is a well-defined process that helps in organizing the software development life cycle, ensuring that the end product is reliable, scalable, and meets user requirements. By following a disciplined approach to software development, the project ensured improved quality control, documentation, and risk management during the software development process.

For the proposed project, an iterative and incremental SDLC approach was followed because of the multi-component nature of the system. The system is an integration of multiple components, such as a web-based user interface, machine learning prediction engine, authentication system, chatbot system, and cloud-based database. The iterative approach helped in developing the system in phases, where each phase of the system development life cycle focused on the implementation and improvement of specific functionalities like user authentication, assessment module, prediction engine, and recommendation system.

During the requirement analysis phase, the objectives of the system were determined, which included early mental health assessment, suggestions, and online support. Functional and non-functional requirements were determined during this phase by analyzing the requirements of the users and the objectives of the system. This phase helped in developing a proper understanding of the scope of the system, the target audience, and the expected results.

Designing phase consisted of the design of architectural design. The designing of front end, back end, and database, together with the implementation of machine learning model design, was carried out in this phase. A friendly front-end design was developed, along with scalable and secure back-end design. Also, communication of API and cloud server deployment was carried out in this phase.

Implementation phase consisted of the implementation of front end and back end using respective technologies. The front-end implementation was carried out using web technology, whereas the back-end implementation was carried out using Python-based technology. Mental disorder prediction machine learning models were developed in this phase. Interaction chat bot, recommendation generation, and admin modules were developed in this phase.

Testing phase consisted of carrying out various kinds of testing, including functional testing, usability testing, and performance testing. These tests were conducted to ensure the correct functioning of modules. Performance-related bugs, security, and performance issues that might be detected during testing phases were corrected. Bugs, security holes, and performance problems were identified and fixed by continuous testing cycles. User acceptance testing was also performed to test usability and overall system functionality.

Finally, the deployment and maintenance phase made sure that the system was accessible to users through web hosting and cloud technology. The system can be updated continuously after deployment through monitoring and maintenance. The continuous SDLC process makes sure that the system stays relevant and functional.

4.4 System Design

The system design of the proposed artificial intelligence-based web-based mental health support system is expected to create an integrated, scalable, and optimal system design that integrates the user interface, machine learning, and database management system. The system design of the proposed system is based on a modular system design that breaks down the system into a series of modules designed to complete a specific function, such as user authentication, data collection, data preprocessing, prediction, recommendation, chatbot, and result visualization.

The modular system design of the proposed system increases system maintainability, scalability and reliability and enables the development of the different system components to be performed independently. The proposed system is also designed with a user-friendly and intuitive frontend system design.

The frontend system design of the proposed system allows users to interact with the system using psychological assessment forms and questionnaires that gather psychological factors associated with stress, anxiety and depression. The proposed system's frontend system design is also able to allow users to interact with chatbots, recommendations and dashboards. It also has a system design that is compatible with multiple devices, such as desktop computers and smartphones, to facilitate easy access to users.

The backend module will take care of the business logic, authentication and machine learning interactions. Once the users have completed the assessment, the backend module takes the information and passes it to the integrated machine learning model for prediction. The model processes the information and creates a classification of the mental health status, which is then automatically returned to the user. The communication between the frontend and the backend module is done using the RESTful APIs.

A cloud-compatible database such as MongoDB or Firebase is used to maintain the profiles of users, encrypted credentials, responses to the assessment test, predictions, chatbot conversation logs, and progress data. The database is designed to support flexibility and scalability to store and retrieve structured and semi-structured data efficiently.

Appropriate indexing and grouping strategies are used to ensure the database is efficient when dealing with a substantial amount of data. The system also has features for dashboard and report generation to monitor mental health progress. The system also includes administration tools to monitor the system and update the system content while maintaining the system user's view.

The system design incorporates security features such as authentication, authorisation, data validation and encryption communications to ensure the integrity of sensitive mental health data. The system provides confidentiality and integrity. The system design ensures that there is a proper integration of all system components, efficient data flow, scalability for future system expansion and stability. The system uses artificial intelligence and new web technologies to build a stable and structured digital mental health assessment and support system.

4.4.1 Data Flow Diagram

The Data Flow Diagram (DFD) is a diagram of the information flow in the proposed web-based mental health support system with AI capabilities. The DFD illustrates the flow of information from one component to another in the system, such as from the user, web interface, processing system, machine learning system, database, chatbot system and the administrator. The DFD is a logical representation of the system without going into the system development life cycle.

At the initial level, the main external entity is the user, who will interact with the system by registering, logging in, and filling out mental health assessment questionnaires. The user will input the answers to the questionnaires using the frontend interface. The input answers will be submitted to the backend server for validation and processing. The backend server will then send the processed information to the machine learning module, which will analyze the answers and produce prediction results of the mental health status in terms of stress, anxiety, or depression levels.

The result of the prediction outcome is then sent back to the backend and stored in a secure way in the database along with the user profiles and history. Then the system gives feedback, specific suggestions and coping strategies based on the outcome. Further, the chatbot component communicates with the user through accepting the user's inquiries and giving instant responses, and also keeps a record of the chat history for continuous and improving communication.

The administrator is another external entity that can access the system monitoring feature, data set, change the questionnaire and manage users from the administrator panel. But the administrator can only view the private data (user data) if given permission by the user, ensuring data privacy.

The Data Flow Diagram illustrates the flow of data from the input to the prediction outcome, recommendation, data storage and monitoring by the administrator.

DFD Level 0 (Context Diagram)

The context diagram or Level 0 Data Flow Diagram gives an overview of the proposed web-based AI mental health support system by modelling it as a single process. The context diagram shows the interaction between the system and the external entities like the user and the administrator, but does not reveal any technical details about the system. The first external entity is the user who inputs data to the system, such as credentials for registration, responses to psychological tests and queries to chatbot. The second external entity is the administrator who monitors the system, data management and content updates.

The system accepts inputs from the user via the web application and processes the inputs based on the back-end logic and machine learning algorithms. The system generates mental health predictions, recommendations and chatbot responses based on the user's assessment. The system helps the user to understand their mental health status and to engage in self-management or seek professional assistance when necessary.

The database stores and retains the processed data, along with the user profiles and history of the mental health assessments. Administrators can access system data and manage system functions while protecting users' privacy. The Level 0 DFD is a basic diagram that shows the flow of data from external entities to the system, making it possible for stakeholders to understand the system boundaries and functionality without knowing how the processing takes place.

Level 1 Data Flow Diagram (DFD)

The Level 1 Data Flow Diagram (DFD) is a more detailed description of the proposed web-based mental health support system that uses AI, which shows the overall process divided into a series of internal processes.

The Level 1 DFD, unlike the Level 0 DFD, is a description of the internal data flow between the most important functional processes of the system, such as managing users, inputting assessment data, processing data, machine learning based prediction, producing recommendations, chatbot interaction, and reporting.

This level of description is helpful in describing how the subsystems of the system work together to offer intelligent mental health support. The first process is user authentication. Here, users sign up and use a password which is stored in an encrypted fashion. Once users are logged in, they can use several features of the system, such as mental health assessment, chatbot, and reporting. Following successful login, the users go to the data collection module to respond to mental health assessment questionnaires.

The data collected is related to psychological symptoms of stress, anxiety and depression. The data then goes to the data processing module, where the data is validated, cleaned, normalised, and formatted. The data is now ready for analysis. The data is then transmitted to the machine learning prediction module. Here, the prediction models predict the current status of the user's mental health, which is categorised into normal, mild, moderate or severe.

The prediction outcome is generated in real-time and passed to the following module to be stored and displayed. Once the prediction outcome is generated, the outcome is securely stored in the database along with user profiles, assessment records and chatbot conversations. This enables users to track their progress and review their previous assessments.

The recommendation component uses the prediction outcome to recommend strategies to manage stress, lifestyle and mental health wellness according to the user's mental health status. Meanwhile, the chatbot interaction module enables users to have chat-based interactions concerning mental health awareness and support. The chatbot is able to respond to user questions automatically, while capturing user interaction data to enhance its intelligence and continuity.

The reporting and visualization component is in charge of retrieving data from the database to generate reports and summary pages that present mental health trends. Users can access their mental health trends and progress via summary pages that encourage self-awareness and engagement. Moreover, the administrator component is responsible for the management of system configuration, user, assessment content and data. Administrators ensure the system works well, enhance the model, and monitor the system usage without compromising user privacy.

4.4.2 Use Case Diagram

The Use Case Diagram of the proposed AI-based web-based mental health support system illustrates the connection between the users of the system and the functionalities of the system. The Use Case Diagram helps in understanding the major actors of the system and their interaction with the system functionalities. The major actors of the proposed system are users and administrators.

The Use Case Diagram helps to understand the system functionalities in an easy way, by showing the actions of the actors and the services provided by the system. The primary actor of the proposed system is the user who interact with the system using the various features. The users can register and login to the system by using the authentication services. Once logged in, the users can answer mental health questionnaires to identify their levels of stress, anxiety and depression.

The system provides users with the prediction and recommendations. They can also chat with the AI-based chatbot to get recommendations, understand how to resolve mental health problems and ask mental health-based questions. The other important stakeholder is the administrator who acts as a system manager.

The administrative tasks include user account management, management of mental health assessment questionnaires, management of system performance, management of datasets and management of updates for machine learning models. The administrators ensure the system is efficient and secure without having to look at the users' sensitive data unnecessarily, thus preserving privacy and confidentiality.

Primary use cases include user registration, user login, submission of mental health assessment, predictions generation, recommendation generation, chatbot interactions, report views, and progress tracking. Use case identifies the function that should be provided by the system in order to meet the needs of the users and facilitate easy access to the mental health services offered.

The relationship among actors and use cases indicates the level of responsibilities that each actor will have with respect to the system functionalities. In general, the Use Case Diagram helps in providing the understanding about the scope, interactions, and functionalities of the system.

4.5 Database Design

The database design for the proposed AI-driven web-based mental health support system is developed to efficiently store, manage, and retrieve data related to user interactions, assessment results, and prediction outcomes. As regards database architecture, the goal will be managing data related to user interactions, user mental health assessments, and the result of prediction generation. Cloud database architectures, such as MongoDB and Firebase, can be considered in this case, as they will allow managing the data in real time.

The database is well organised into various collections or tables for different purposes. These are user data, assessment data, prediction data, chatbot conversation data and administrative data. Each data is uniquely identified with secure identifiers in order to maintain data integrity and avoid data duplication. Data relations are established via reference identifiers, so it is easy to link user data with their assessment data, recommendations and mental health journey.

Data integrity and security is also critical in the design of the database. Entry rules are put in place to ensure that only valid data is entered into the system. Access rules are also used to ensure that there is no unauthorised access to the system, ensuring the privacy of the mental health data. Encrypted algorithms are also used for data storage and transmission to ensure security and privacy and to prevent breaches. There is also a mechanism in place to backup and recover data to ensure the data is not lost. The solution also ensures the database is scalable.

The adaptability of NoSQL databases means that it is easy to add new features, such as analytics, mood tracking or intervention, without altering the database design. Indexing and efficient queries also allow data to be easily and quickly searched for dashboards, reports and AI insights. The database design ensures efficient, scalable and secure data management, which is important for the successful operation of the mental health support system and for monitoring well-being over time.

CHAPTER 5: IMPLEMENTATION

5.1 Tools and Technologies Used

The proposed AI-based system will use web technology, machine learning, and cloud computing. As described earlier, the implementation of the suggested solution will require a multistage approach encompassing frontend design, backend design, AI, databases, and deployment of applications.

To implement frontend technologies, tools such as HTML, CSS, and JavaScript will be used. The creation of frontend technologies will enable designing forms, dashboards, and graphs necessary for registering users, diagnosing their conditions, and communicating with the bot. One of the advantages of web applications is the ability to access the system through various gadgets without installing additional software.

Backend design will consist of authenticating and managing data, utilizing machine learning models, and ensuring intercommunication between modules. To implement server-side programming, a language like Python will be used. Moreover, it will be required to design RESTful API, which will allow frontend, machine learning, and database modules to communicate with each other.

Machine learning is a critical component in the development of the system. The predictive module is responsible for the analysis of user responses to mental health questionnaires and the prediction of mental health conditions such as stress, anxiety, and depression levels. The analysis can be carried out using algorithms such as Logistic Regression, Random Forest, or Neural Networks depending on the required level of accuracy. Data preprocessing operations such as cleaning, normalization, and feature selection are carried out prior to training the model to improve the accuracy of predictions. The trained model is then used in the backend system to make real-time predictions.

Table 5.1.1 Tools and Technologies used

Category	**Tool / Technology**	**Purpose**
Programming language	JavaScript (Node.js)	Core development and implementation of vehicle detection system
Frontend Technologies	HTML, CSS, JavaScript	Building responsive user interface and web interaction
Database	MongoDB / Firebase	Storing user data, assessments, and prediction history
Dataset Source	Public mental health datasets / Custom survey data	Training and validating prediction models
Database	MongoDB	Storing the parking slots status and historical data.
Backend Architecture	Express.js	Lightweight framework for building RESTful APIs and managing routing and middleware.
Authentication	JWT (JSON Web Token)	Secure access control (when web dashboard is done)
Security	bcrypt.js	Password hashing and encryption for secure user authentication.

5.2 Dataset Description

The data used for the development of the proposed AI-driven web-based mental health support system is based on the emotional, behavioural and psychological information of the users. This dataset mainly includes responses to mental health surveys, levels of moods, levels of stress, symptoms of anxiety, sleep patterns, lifestyle choices, and social interactions. These are very helpful to identify potential mental health issues, including stress, anxiety or depression. The dataset is used to train the machine learning models for predicting the user's mental health issue. The dataset includes a list of variables that represent different mental health issues of a person.

These variables are divided into demographic variables (age group, and occupation), emotional responses, sleep habits, work-life balance variables, social interaction variables, and perceived stress level. The rows in the dataset correspond to different users and the target variable represents the mental health issue predicted for the user. The dataset needs to be appropriately labeled to train the machine learning model. The data is preprocessed before the model is trained to increase the data quality and consistency.

Data preprocessing techniques include dealing with missing data, removal of duplicates, normalization of continuous data, and encoding of categorical data. Data preprocessing helps the machine learning models learn from the data effectively. Other techniques like feature selection can be used to select the most important features affecting mental health, thereby increasing the efficiency of the model and preventing overfitting. We split the data into training and testing samples to assess the model's performance.

This dataset is used for creating predictive models, while the testing dataset is used to compute evaluation metrics like accuracy, precision, recall and F1-score. This helps ensure the model will perform well with new data not used in the training process. The model can be retrained and data can be updated to get better predictions and to be in line with the recent trends in human behaviour.. The data is handled in accordance to the code of ethics. Since the data is related to mental health, confidentiality and privacy is taken seriously. Personal data is deidentified when required and secure data storage measures are taken to keep data safe from any kind of breach. Consent is also taken from the users prior to data being uploaded to ensure highest ethical standards are maintained. The data can be obtained from publicly available data on mental health data sources, psychological tests and surveys collected anonymously on the website. The data obtained from various sources can be used to obtain more representative models. Sampling techniques such as oversampling, undersampling or using synthetic data can be used to compensate the data, ensuring all types of mental health conditions are represented.

Time series or temporal data can also be used to understand the changes in the user's mental health condition over time. Time-series data helps the system to understand the trends, red flags and progress of the user. This helps in enhancing the system's capability in tracking the user's mental health over time rather than at an instant. Exploratory data analysis is carried out using data visualization techniques. The representation of data in form of graphs, statistics and distribution graphs helps in the analysis of data and the selection of optimal features. The description of data source, data preprocessing, features and limitations is documented well to ensure.

CHAPTER 6: TESTING AND MAINTENANCE

6.1 Testing Overview

Testing is one of the important stages of the development process of the proposed Mental Health Monitoring and Support System (MHMSS) to validate that the system is working properly, securely and efficiently. The key focus of testing is to detect the errors in the system, to validate the functional requirements of the system and to make sure that the individual modules of the system are working correctly.

This phase ensures that the user interface, APIs, AI chatbot, authentication mechanism, and the database are functioning as expected. The testing process that has been done includes unit testing, integration testing, system testing, security testing and user acceptance testing. Unit testing involves the testing of different components of the system, which includes user authentication, mood entry, chatbot, and appointment booking APIs. Integration testing involves the testing of various components of the MERN stack technology which includes React frontend, Node.js server, Express APIs, MongoDB database, and AI systems. System testing includes the validation of the complete system based on functional and non-functional requirements.

6.2 Types of Testing Performed

The following testing techniques were applied during system validation:

6.2.1 Functional Testing

Functional testing was carried out to verify that the functionalities of the mental health system are working as planned. It was done to ensure the functionality of basic features like user registration and login, mood tracking, chat with the AI chatbot, appointment registration and the visualisation of reports. We performed this by testing each module with certain inputs and outputs to see if the system's output is the same as the expected output.

The authentication module was tested for secure login and access control. The mood-tracking APIs were tested to confirm the functionality of storing and retrieving mood data. The AI chatbot was tested for relevant and contextual responses. Error handling has also been tested to ensure proper functioning of the system with incorrect or invalid inputs.

6.2.2 Integration Testing

Integration testing has been performed to ensure the proper integration of different parts of the system. As the system is developed on MERN stack and AI integration, it is important to check the interaction between the frontend and backend API and AI API's.

The interaction of React frontend and Express backend APIs has been verified. The interaction of the backend APIs and MongoDB has also been checked for storing and fetching data. Integration of AI chatbot and external APIs has also been tested for the evaluation of moods and generating smart responses.

6.2.3 Security Testing

The security testing was also conducted to secure user data such as moods, chatbot conversations, and user information. The application is about handling very sensitve mental health data so authentication and encryption were done. The login process was tested for JWT to ensure that only logged in users access to protected routes.

The password hashing functions were tested to ensure that the passwords are stored securely. The application was tested for role-based access for users and admins. Security testing also involves testing for other common vulnerabilities such as accessing restricted resources and inputs.

6.2.4 Performance Testing

System performance testing was conducted to ensure that the system can handle various levels of traffic. This was to verify the system is capable of handling many users efficiently and in a timely manner.

The response times of the API calls were tested. The efficiency of MongoDB queries with a large data set was tested. The scalability of the AI chatbot was also tested to check for any lagging in real-time communication. The system is identified to be performing well under heavy load.

6.2.5 User Acceptance Testing

Testing was done to check the acceptability of the software from the user's point of view. A group of users was asked to use the system and test its features for mood tracking, chatting with the chatbot and viewing the dashboard.

The results of UAT reflected that it is easy to use, user-friendly and that the AI chatbot provides a lot of help. Minor improvements to the user interface were made to enhance the user experience.

6.3 Test Case Design

Designing test cases was One of the factors that helped in the proper validation of the system was designing test cases. The test case included test ID, test case description, input, expected output, actual output and test result.

The test cases were designed for the key modules of the system like user authentication, mood management, chatbot, appointment scheduling and database operations. Valid and invalid test cases were executed to test the input validation and error handling.

The test cases ensured the flow of data from frontend to the backend and to the database services was not blocked at the boundaries of the modules. Test cases helped in the early identification of bugs and the debugging process was faster.

Table 6.3.1 Sample Test Cases

Test Case Id	Module	Test Description	Expected Result	Status
TC-01	User Registration	Register new user with valid details	User account created successfully and stored in database	Pass
TC-02	Authentication	Login using valid credentials	User successfully logged in and redirected to dashboard	Pass
TC-03	Authentication	Login with incorrect password	Access denied with error message	Pass
TC-04	Quiz Module	Attempt mental health assessment quiz	Quiz loads properly and accepts responses	Pass
TC-05	Result Analysis	Submit quiz and calculate score	Accurate mental health score generated with category	Pass
TC-06	Recommendation System	Generate suggestions based on quiz result	Personalized tips and coping strategies displayed	Pass
TC-07	Chatbot Module	Ask mental health-related query to chatbot	Relevant and supportive response generated	Pass

6.4 Test Environment

The testing environment was designed to closely simulate real-world judicial usage.

Table 6.4.1 Test Environment Configuration

Component	Specification
Frontend	React.js
Backend	Node.js with Express
AI Model	OpenAI API / Gemini API for AI therapist chatbot and mood analysis
Database	MongoDB (NoSQL)
Authentication	JWT based secure authentication
Deployment	Cloud-based
System Development Tools	VS Code, MongoDB Atlas, GitHub
User Simulated	Admin, Registered User, Guest User

6.5 Test Results and Observations

The mental health monitoring system proposed is stable, secure, and reliable. In the functional testing stage, we observed that all modules, including authentication, mood monitoring, communication with the AI chatbot, and making appointments worked effectively. The data inputted by the users was stored and retrieved by the database system without any loss of data.

In the integration testing stage, the collaboration among front-end and back-end APIs, database and AI systems worked flawlessly. The performance test results indicate that the response rate of the system to many users was impressive. In the security testing stage, we noted that the system had robust authentication and encryption capabilities for the users' data. Some bugs in the system were fixed to enhance the stability of the system.

The mental health monitoring system achieved our goal of providing smart mental health care, data handling, and a friendly user interface successfully. However, the system needs improvement in scalability, AI chatbot accuracy and features addition.

6.6 Maintenance Strategy

The maintenance philosophy of the proposed Mental Health Monitoring and AI Support System is maintaining system availability, scalability, security, and performance. Considering that the system deals with sensitive emotions, interacts with the AI and monitors user behavior, the maintenance process should be proactive and well organized. The key aspects of maintenance will include stability, security, AI development, and the compatibility of the system with emerging technologies.

Corrective maintenance means fixing the errors and faults that arise during operation of a system. It will involve addressing problems with the backend, such as errors in the API, session errors, authentication errors, and errors in the interface. Logs from the Node.js server will be used to monitor the performance and address errors with chatbot responses and mood tracking.

Adaptive maintenance is carried out to ensure that the system is compatible with the latest environment and technology. Since the system runs on the web, adaptation will involve ensuring that it is compatible with the latest browsers and mobile phones. On the backend side, it will involve updating Node.js and Express.js and upgrading the MongoDB schema.

Adaptive maintenance also involves making sure that the system is compliant with the latest data protection and cloud deployment strategies. Perfective maintenance is about enhancing the performance, usability, and quality of features based on user feedback.

Perfective maintenance will involve improving chatbot intelligence to give more relevant responses, improving the dashboard mood trend views, improving API response time, and improving UI/UX designs. Feedback from testing is used to implement improvements in user satisfaction such as easier navigation, faster load times and easier mood tracking.

Preventive maintenance is designed to prevent system failures in the future. Regular backups are made of the database to prevent data loss. Regular code reviews and vulnerability scans are performed to detect vulnerabilities. Load testing is also done to ensure the system is able to handle a large volume of users. Preventative maintenance also involves keeping libraries and dependencies up-to-date to prevent compatibility and security problems.

AI model maintenance and integration is vital in the inclusion of AI chatbots. The AI components are checked on a regular basis to ensure the responses are appropriate. The changes may involve the fine-tuning of response prompts, sentiment detection, and updates to the latest API of the AI models when available. Chats and feedback are used to provide better context and avoid incorrect or non-specific responses.

Security and privacy are also important as the website deals with personal and sensitive information. Improvements are made to the authentication system such as JWT and password hashing libraries. The role-based access control is also investigated to avoid non-authorised access. The data is also encrypted and the APIs are secure to ensure privacy of the users. The vulnerability issues are patched as soon as they are detected to adhere to ethical practices in data management.

Monitoring and tracking the system is also done to ensure the system is efficient. Monitoring tools are employed to monitor server uptime, API response times and database performance. Log monitoring is used to detect irregularities, such as spikes in traffic and failed requests. Administrators are notified of any abnormal activity..

6.7 Conclusion of Testing and Maintenance

The testing and maintenance phase is a critical phase in ensuring the proposed Mental Health Monitoring and AI Support System is reliable, efficient, secure and sustainable. The system was tested using systematic approaches to evaluate the system on its major components, which include user login, mood tracking, interactions with the chatbot, database management and API interactions. The result of the testing has been assuring that the system is working as expected and it is providing robust and efficient services to end-users.

While testing the system, the system's modules (such as user interface, APIs and AI) were tested using functional, integration, security and performance testing approaches. Functional testing was done to ensure that key features of the system, including user registration, login, mood tracking, chat with the chatbot, and the display of the dashboard were working as intended. Integration testing was carried out to ensure the interaction between the React front-end, Node.js back-end, MongoDB database and AI APIs was seamless and not resulting in data loss or de-synchronisation.

It also ensured that the data testing and cleaning methods were implemented properly to ensure the proper storage and retrieval of the user's mood data. The database methods were correctly implemented and scalable, and the reporting methods were showing the correct mood analysis and activity reports. Users faced some minor usability and response time issues that were fixed by fine-tuning the user interface. Also, a correct maintenance plan was prepared to ensure proper functioning of the system in the future. Corrective maintenance involves bugs and errors at runtime, usability problems, and server errors to ensure the system is fully functional.

Adaptive maintenance involves making sure that the system is up-to-date with the current technology, browser, and API. Perfective maintenance includes the enhancement of the intelligence, user interface, and performance of the chatbot based on feedbacks and testing.

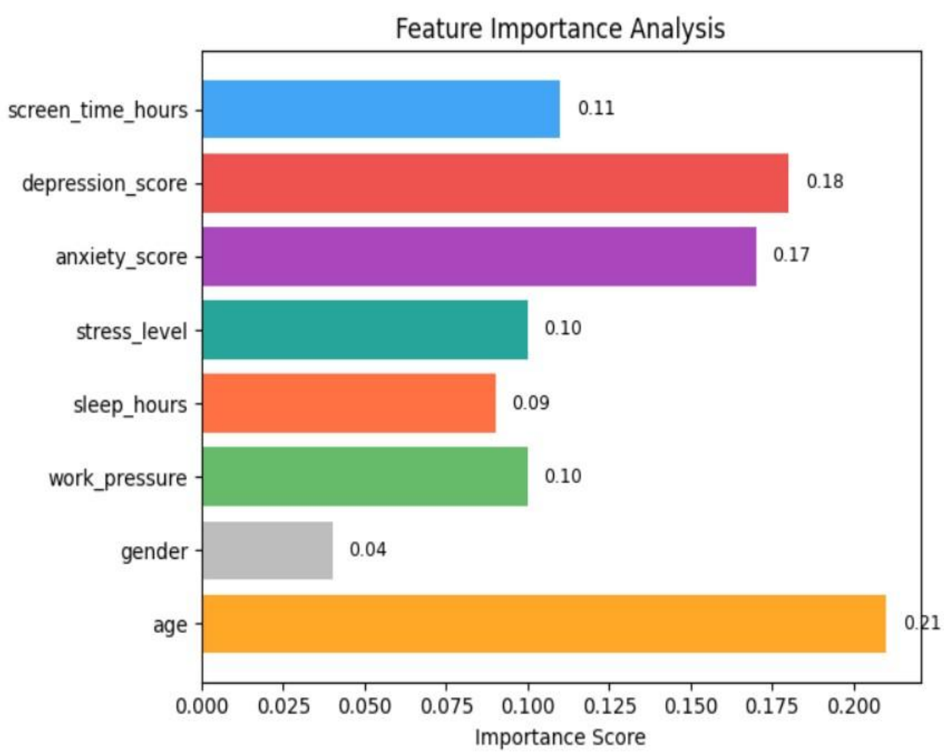
Preventive maintenance involves the backup, upgrade and monitoring of the system to prevent system failure. Particular attention was given to how to maintain the AI components and the privacy of the system. The system's testing and maintenance phases demonstrate that the proposed system is robust, secure and scalable for deployment.

The planned testing phase of the system ensured the system works and the planned maintenance phase of the system allows it to be improved. The two phases together make a constant and reliable platform for the practical implementation of a digital mental health system for early emotional recognition and AI-based mental health support in the future.

CHAPTER 7: RESULTS AND DISCUSSION

7.1 Presentation of Results

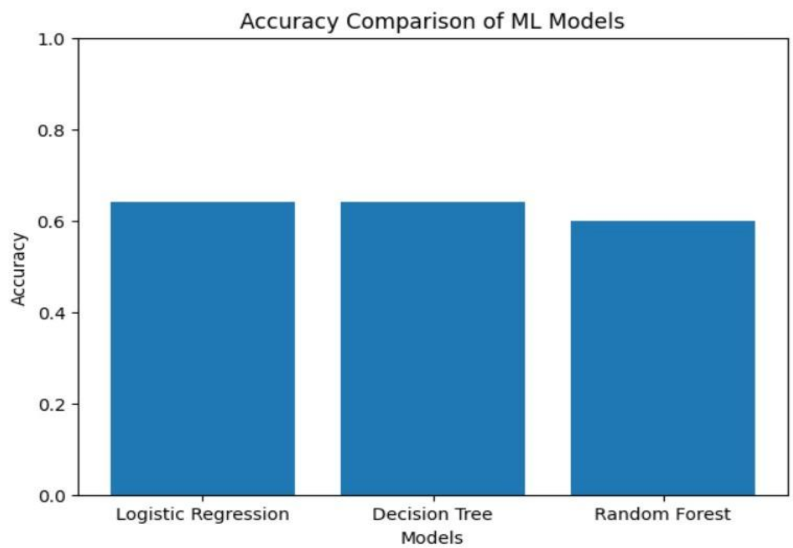
In the process for the application of the implementation for the online mental health support system, there have been results for the identification of the factors that affect the mental health conditions for the users within the system. This has been carried out through the evaluation for the system's machine learning model for the identification of the important factors for the attributes that affect the system users for the resultant mental health conditions for the system users. There are results for the correctness of the system's predictions in relation to the psychological principles.



Model Training and Quiz Design:

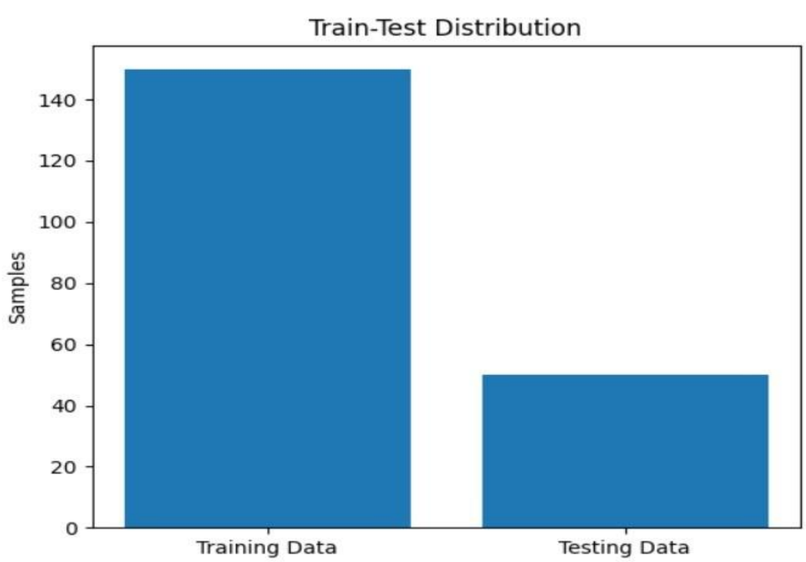
In mental health status prediction, there have been three supervised learning models used. The accuracy of these models has been calculated. The results are as follows: Model Accuracy Logistic Regression 0.64 Decision Tree 0.64 Random Forest 0.60 Logistic Regression models are able to handle the linear part of mental health parameters.

Decision Tree models are useful in rule generation, which is very much appropriate for use in quizzing systems. Random Forest models are capable but require a large amount of data to demonstrate their capability.



Train-Test Distribution Analysis:

The dataset was split using the 80:20 split ratio for training and testing purposes. This entails that the dataset will have 150 samples for training purposes and 50 for testing. This split of the data ensures that the network gets an optimal amount of data for training. This is important for the network to generalize well in real-world applications of the model in mental health applications.



7.2 Performance Evaluation

The performance of the proposed AI-based Mental Health Monitoring and Support System was evaluated by using a combination of machine learning evaluation metrics and system performance criteria. The objective of the performance evaluation was to investigate the accuracy of predictions, reliability of models, effectiveness of the chatbot system, and systemlevel stability for the identification of emotional risk patterns and intelligent mental health support.

1. Evaluation Metrics

To evaluate the proposed mental health model effectively, a combination of classification metrics was used. Accuracy was used to determine the overall percentage of correct predictions made by the mental health classification model. Precision was used to determine the accuracy of the system in identifying users at risk without generating a high number of false positives. Recall (Sensitivity) was used to determine the system's ability to make correct predictions about actual emotional distress cases, which is of utmost importance in mental health systems. The F1-score was used to determine a balanced measure of precision and recall, which is of utmost importance in class-imbalanced problems.

2. Model Comparison and Accuracy Analysis

Several machine learning models were tested in the development phase of the project such as Logistic Regression, K-Nearest Neighbors (KNN) Random Forest and Gradient Boosting. Our experimentation showed that ensemble models such as Random Forest and Gradient Boosting were better at detecting complex patterns in the emotional data. These algorithms were more effective in capturing the non-linear patterns in the mood and behavioral data collected from mood logging and chatbot conversations. Models such as Logistic Regression were moderately good but not when there was an overlap between emotional features or the data were imbalanced.

3. Feature Importance Analysis

The feature importance figure shows the contribution of individual input features towards predicting mental health risk levels. It is evident from this analysis that age was the most important feature, followed by emotional features, such as depression score and anxiety score, which were crucial in determining the outcome. Other features like screen time, sleep time and work pressure were moderately important, which means that they are secondary factors affecting mental health. The least important was gender. This study confirms that different psychological features and lifestyle features have a collective influence on mental health status and supports the creation.

4. Training and Validation Performance

Learning curves on training and validation datasets were used to analyze the model learning process. The accuracy of the training set continued increasing, while the accuracy of the validation set did not show any variations with each iteration. It was evident from the loss curves that oscillations occurred in the smallest amounts; this means that the simpler models were experiencing slight underfitting as a result of their inability to capture the hidden emotional complexity. The reason for using ensemble models and AI to improve prediction accuracy of the final system was this.

5. Results of Confusion Matrix Analysis

According to the results of confusion matrices, there was a slight bias of the model towards frequent emotions and some degree of confusion with minority classes related to mental health risks. The problem of classification between similar emotional states such as mild anxiety and moderate stress was noted. Based on the findings obtained, the need to improve feature selection, balance the classes and retrain the model with real-life mood data extracted from the mobile application became evident. According to the classification report, precision and recall values for minority emotional classes were lower compared to dominant ones.

6. System-Level Performance Assessment

Besides the model's performance, the system's performance was also evaluated. The optimal API response time was achieved through the use of Node.js and Express.js, which made it possible to interact between the dashboard and the AI system. Optimal storage of data related to the mood log, chat, and user analytics was obtained by utilizing MongoDB. There were optimal responses and interaction when integrating the AI chatbot. The optimal insights were also achieved during the reporting and visualization processes. Security measures such as JWT token and data encryption were optimal in protecting the sensitive emotional data.

7.3 Key Findings

Our design, development and evaluation of the proposed AI-based Mental Health Monitoring and Support System resulted in a set of key observations that relate to model performance, effectiveness of AI, system robustness and future improvements. These insights were realised through experimental testing and analysis of emotional and behavioural data collected from mood tracking and chatbot interactions.

A key finding was that the ensemble machine learning models (Random Forest and Gradient Boosting) performed far better than the simpler logistic regression models. Ensemble models were able to perform better in overall accuracy and were more capable of modelling complex and non-linear interactions between emotional indicators, behavioral features and lifestyle factors. These models were better suited to classify the emotional risk levels and were well suited for application in a practical mental health risk prediction system.

The evaluation analysis indicated that some models had a bias towards the majority emotional risk classes. The results of the confusion matrix and classification report indicated that the recall and F1-scores of the minority emotional risk classes were low. In some cases, the high-risk emotional states were misclassified as moderate and low-risk emotional states.

This highlights the importance of applying data balancing techniques such as resampling, classweighting or synthetic data generation to improve the system's ability to detect the minority emotional states in future iterations of the system. A review of the training and validation loss curves indicated an underfitting of the simple models. The loss function showed little improvement and the accuracy on the validation data was flat across the number of epochs.

The findings clearly demonstrate the importance of advanced feature engineering and tuning, and ensemble modeling, in improving the classification accuracy of the models A key factor in the analysis of the system is the importance of longitudinal monitoring of mental health. Mental health monitoring over time was very successful.

Mood monitoring and having conversations with the AI chatbot increased the information obtained when compared to single assessments. The trend analysis helped in detecting the early onset of mood changes and also helped the user to understand his/her behaviour patterns. A system test showed the stability and scalability of the MERN stack system. The Node.js and Express.js based system had the capability to manage a large number of API calls, and the MongoDB database allowed efficient storage of data from mood logs and chats in a scalable manner.

The testing of the system in the cloud environment showed stable system performance in a multiuser environment, which confirmed the stability of the modular client-server-based system. Security is important for the system due to the emotional data involved. JWT-based authentication, secure communication and role-based access control ensured data security and privacy.

Safely storing data and following good data practices increased user confidence and data compliance readiness. This finding supports the need for security systems in the deployment of real-world AI-based mental health systems. The assessment also revealed some potential improvements in the future.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The purpose of this study was to address the growing issue of mental health monitoring by creating and deploying an AI-based digital mental health system. With the growing exposure of people, especially students and young adults, to stress, anxiety, and emotional imbalances, there is a strong need for accessible, continuous and digital mental health support solutions.

This system offers an answer to this challenge by integrating mood monitoring, conversational AI support and machine learning-based mental health prediction in a web-based system. The developed system offers continuous monitoring of emotions through daily mood tracking and chat conversations with an AI-powered chatbot, unlike most conventional mental health practices that are mainly based on intermittent clinical consultations.

This allows the user to track his/her mental health status over time and get insights about the emotional patterns and triggers. Real-time monitoring of mental health status increases users' awareness of their mental health status and enables early recognition of potential mental health issues, making the system suitable for the prevention of mental health issues rather than the treatment of mental health problems.

Technically, the system has successfully combined a full-stack MERN (MongoDB, Express.js, React, Node.js) web application with smart predictions. The React frontend offers a user-friendly and dynamic interface, and the Node.js and Express.js backend handles APIs, authentication and data manipulation. MongoDB offers scalable data storage for mood logs, chat logs and prediction results.

The use of chatbot APIs and machine learning models for AI integration adds value to the system in real-time emotional recommendations. Privacy and security is a top priority. Use of JWT-based authentication, hashed passwords and secure API calls and responses protects user privacy. Responsible data management and access control also enhance trustworthiness and credibility, crucial in the context of mental health. The findings indicated the system is effective in terms of usability, accuracy and scalability.

Machine learning ensemble models exhibited good performance in classifying emotional risk, and the AI chatbot increased user engagement and accessibility. The system was also able to manage multiple simultaneous users with consistent performance, demonstrating the modular system's validity. Finally, this research has shown that AI-powered digital platforms can play a vital role in promoting mental health, detecting mental illnesses and monitoring emotional states.

The system developed in this study is a scalable, secure and easy to use mental health support system that bridges the gap between emotions and smart digital support. The system will contribute to the development of preventive digital mental health care.

8.2 Future Scope

While the AI-based mental health monitoring system proposed in this work meets all of its key requirements, there is still room for improvement to enhance its intelligence, scalability and efficiency. A major area for improvement is the use of advanced deep learning methods to improve the prediction accuracy.

The system can also benefit from the use of more intelligent conversational AI. Emotionally intelligent AI-based chatbots can be used in the future to improve the system that can respond to the users' emotions, have a memory and engage in therapy conversations. Improvements in Natural Language Processing will help the system interpret the emotions of the users in text format.

The next area of future work is privacy-preserving AI. Using techniques such as federated learning or on-device AI processing could be used to allow the model to learn from user data without sending the data to a remote server. This will boost user confidence and will support the recent international data protection laws.

Another potential future development is the use of multimodal data such as integration with wearable devices, sleep monitoring, physical activity monitoring, or voice emotion monitoring. This could potentially lead to more accurate predictions, as well as a broader understanding of mental health. Enhancements in scalability could potentially enable the system to be implemented in a broader context such as education institutions, workplaces, or telehealth.

Features such as role-based dashboards for mental health counselors, real-time data analytics, and mental health trend analysis could potentially make it possible to scale the system. Another aspect of improvement is related with the creation of culturally adaptive and multilingual AI models.

The ability to support multiple languages and provide cultural recommendations can enhance the system's inclusivity and accessibility. Culturally sensitive mental health resources and recommendations can enhance global adoption. Lastly, retraining the model with data gathered from the application can enhance the performance of the model. The use of reinforcement learning and adaptive personalization approaches can help the system become a fully autonomous mental health coach that provides personalised support.

Overall, future research on increasing AI intelligence, privacy-preserving learning, multimodal analytics and global scalability can enhance the effectiveness of online mental health interventions. This will also highlight the role of AI-based systems in the prevention of mental health, and assist in creating more personalised, accessible and ethical mental health solutions.

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Principal, SSCBS
University of Delhi,
New Delhi, India

Prof. Prabhat Kumar
General Chair
National Institute of
Technology, Patna, India

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Technical Program Chair
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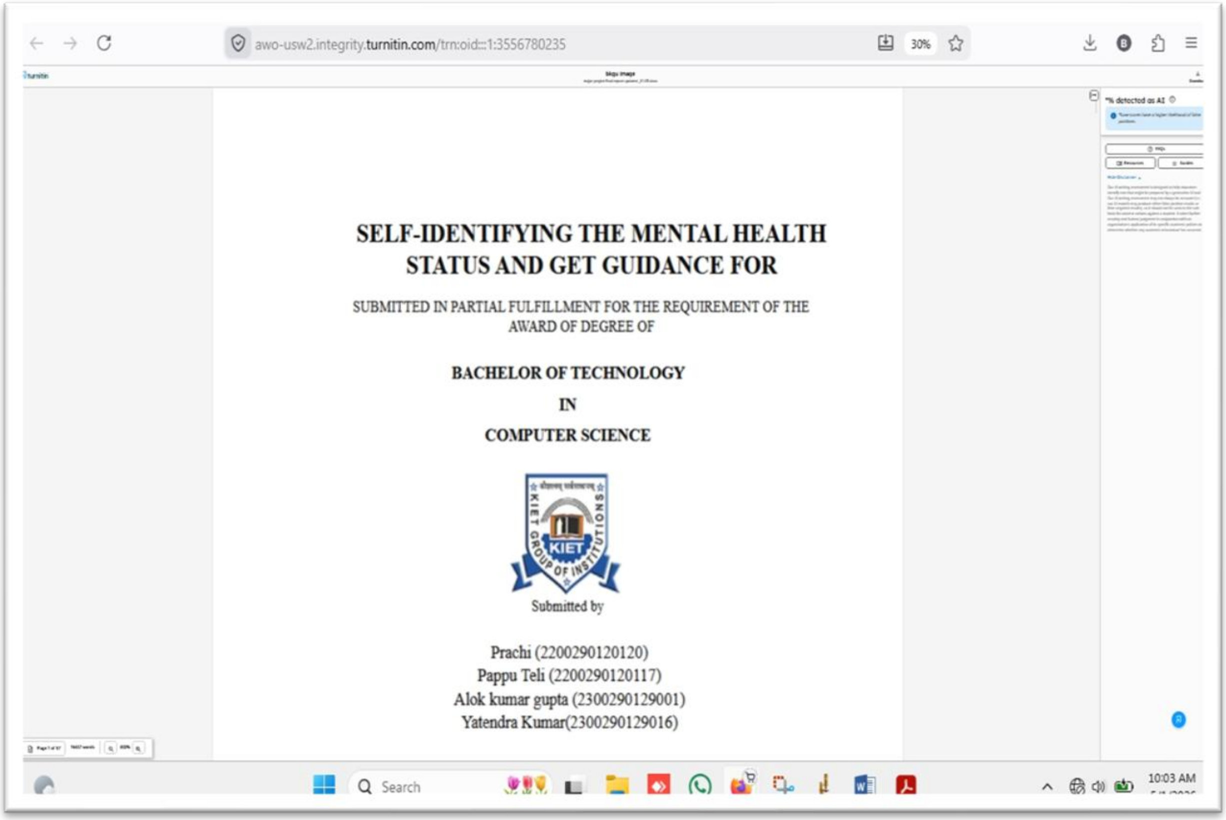
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SELF-IDENTIFYING THE MENTAL HEALTH STATUS AND GET GUIDANCE FOR

SUBMITTED IN PARTIAL FULFILLMENT FOR THE REQUIREMENT OF THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE

Submitted by

Prachi (2200290120120)
Pappu Teli (2200290120117)
Alok kumar gupta (2300290129001)
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